

III

Isostatic Arches

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III.1. INTRODUCTION

In architectural terms, an arch is a curved construction whose two ends will rest on solid points. An arch is an assembly of stone, rubble, brick, concrete, steel or wood, intended to cross a more or less large space by means of a curve (or by the meeting of two curved elements).

The Romans were the first to use arch as main elements for structure (bridges and aqueduct constructions), these arches were essentially semi-circular in shape. A compressive force is developed in these elements, which are built with stones or segments (joined together with mortar). In the middle Ages, their pointed tops, subsequently used intensively in the 19th century, distinguished very detailed Gothic arch. In the 18th century, masonry arch were used to support bridges, several of them surviving to the present. The Ottomans also built their floors and lintels with arch...etc. the historic town of the Casbah is a vivid witness. Today, arches are usually made from steel or prestressed concrete and can withstand very significant tensions and compressive forces, we cite the example of the famous Sydney Harbor Bridge, or arches take up the weight of the deck.

In works, the arch can be placed above or below Fig. (III.1).

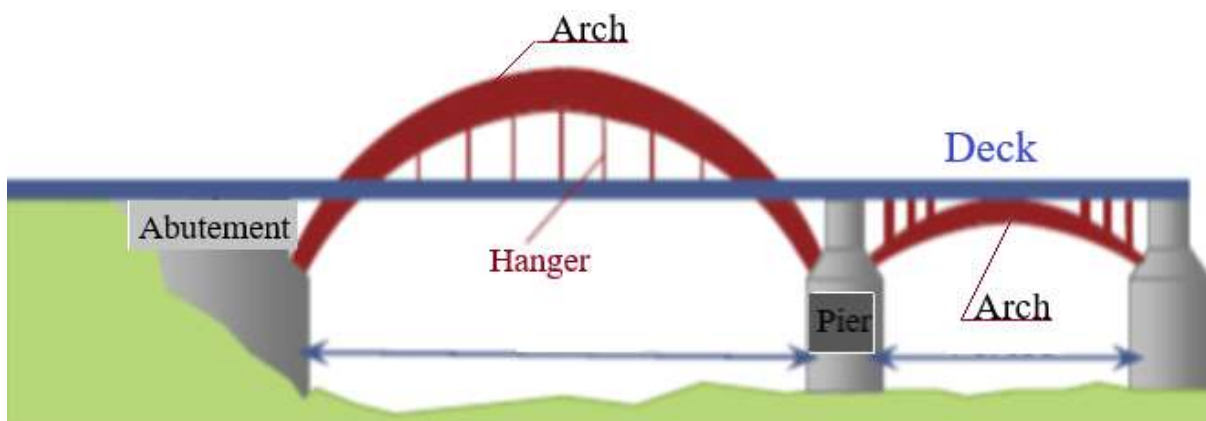


Figure (III.1): Arch placement: right, arch from above with hanging apron, left, arch from below

Arches are constructed in various shapes. Its different parts are linear or curvilinear, but generally fall into two categories:

1. Statically determined tri-articulated arches (isostatic systems).
2. Bi-articulated arches, bi-articulated arches with tie rod and 'bi-recessed or fixed arches and which are statically indeterminate (hyperstatic systems).

Generally, concrete arches are designed hyperstatic. However, the isostatic steel design remains quite widespread, which is the subject of what follows in this chapter.

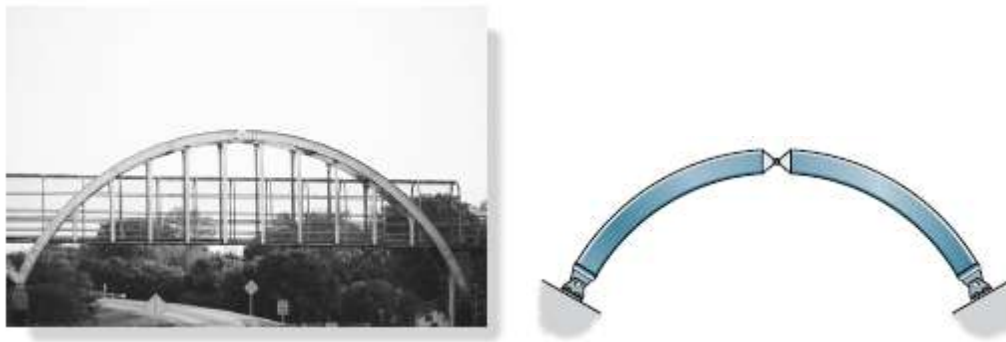


Figure (III.2a): (Left) Steel Pedestrian Bridge suspended from a tri-articulated arch, (right) static diagram of the tri-articulated arch,

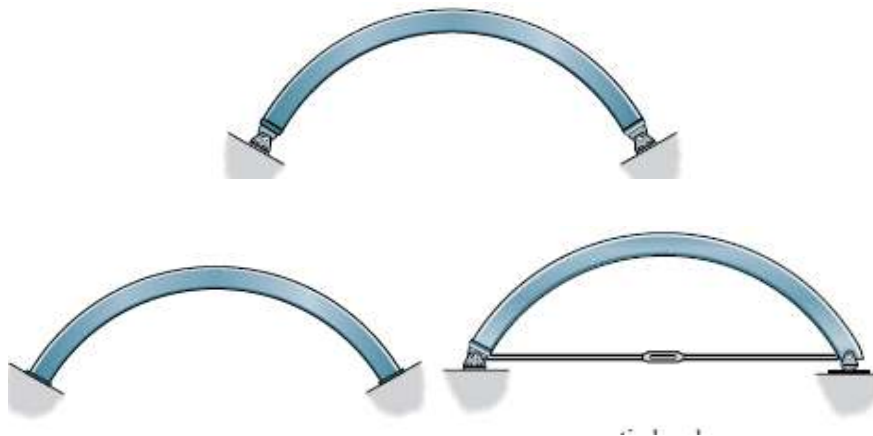


Figure (III.2b): (top) Bi-articulated arch, (bottom left) Fixed arch, (bottom right) Arch with tie rod.

III.2. LINEAR ARCHES

There is a direct relationship between a perfectly flexible cable and the arc. Let the linear cable in (Fig. III.3a) be subjected to two point loads w_1, w_2 , articulated in C and D and double-supported in A and B. If we invert this cable (Fig. III.3b), the magnitude of the internal forces developed remains the same, but corresponds to compression instead of tension, in this case it is the linear arc.

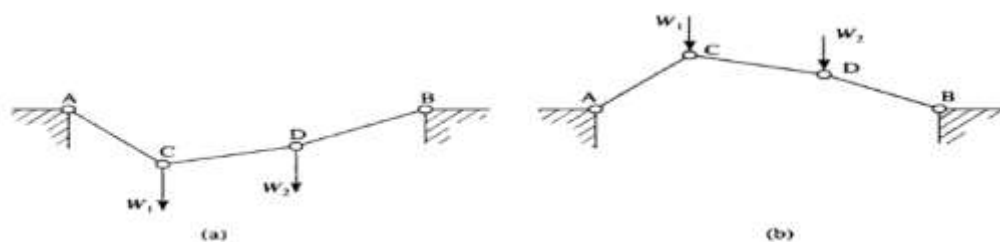


Figure (III.3): (a) Equivalence Linear cable, (b) Linear arc.

III.2.1. Similarity between Linear Arch and Linear Cable

From the above we can deduce that there is a strong similarity between the cable and the arch, in fact:

1. Only the axial compressive force remains in a linear arch, the shear force and the bending moment are zero (linear cable and linear arch).
2. The internal force is purely axial (compression) for a parabolic arch, subjected to loads uniformly distributed horizontally (the shear force and the bending moment are zero).

III.2.2. Difference between Linear Arch and Linear Cable

There is a major difference between the cable and the arch. We know that when moving from a given point loading system to another the cable simply changes geometry. On the other hand, **a slight increase in w_1, w_2 can cause the arch to collapse. We can replace the C (or D) joint with a rigid node to avoid collapse**, the internal forces in the segments of the arch remain unchanged (the geometry is the same), but the arch is more stable. The arch becomes tri-articulated.

III.3 TRI-ARTICULATED ARCHES

Consider the tri-articulated arch below (Fig. III.4a, b), the third hinge (articulation) is located at the top of the arch and the supports are located either at the same level or at different levels. The analysis of this arch is reduced to the global equilibrium and the equilibrium by part (by separating the two members). The unknowns of the problem are on the one hand the support reactions at A and B. On the other hand, they are the internal forces, 'normal force, shear force and bending moment' depending on the type of loading.

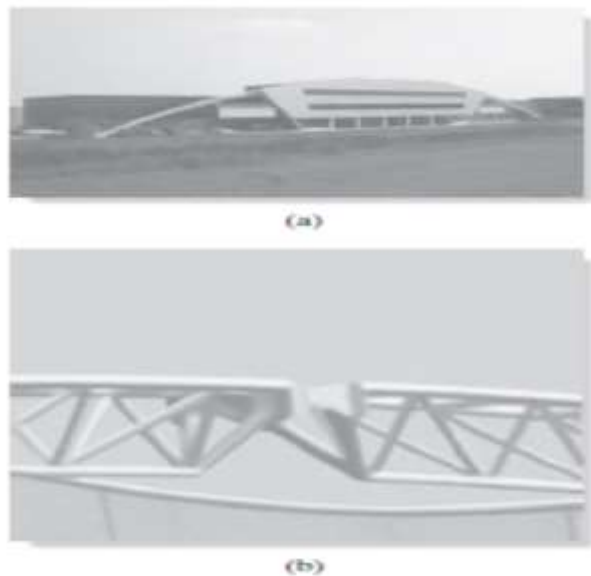


Figure (III.4): **(a)** Tri-articulated Arch Trellis used to take up part of the weight of a roof, **(b)** photo of the detail showing the articulation at the top of the arch.

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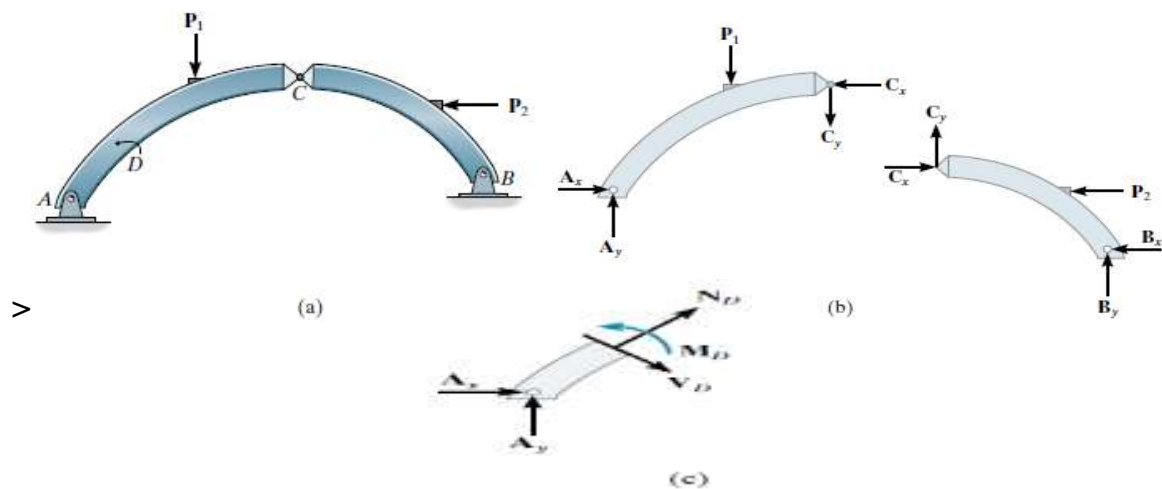


Figure (III.5): **(a)** Tri-articulated arch in A, B and C, **(b)** the six unknowns of the problem: the support reactions ' A_x, A_y, B_x, B_y ', the internal forces at point C ' C_x, C_y ', **(c)** the equilibrium of part AD, internal forces at point 'D' (normal force, shear force and bending moment).

III.3.1. Tri-articulated arch bending moment diagram

The complete bending moment diagram of the arch can be considered as the sum of a bending moment diagram of the simply supported beam under vertical forces (vertical loads and vertical support reactions) given in fig. (III.6b) and a diagram of the bending moment of the arch (in which the diagram of the arch has the shape of the arch itself), since its ordinates are equal to a constant (reaction of horizontal support) multiplied by 'y', given in fig. (III.6c). The two bending moment diagrams can be superimposed as shown in fig. (III.6d), to give the complete bending moment diagram for the arch. Let us take the example below.

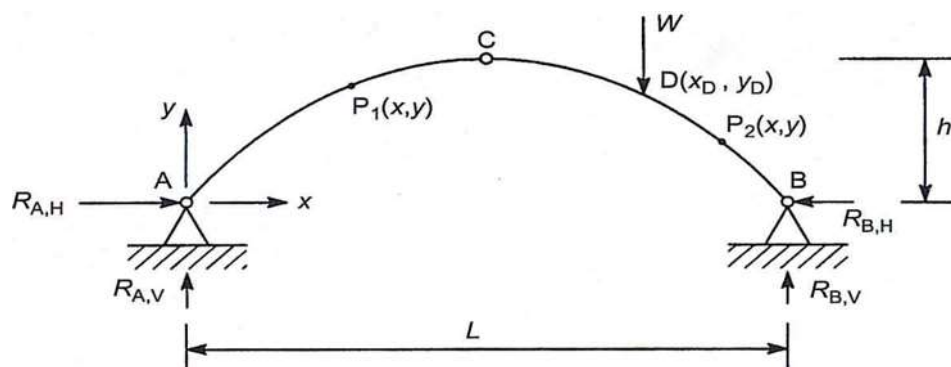


Figure (III.6a): Example: Determination of the moment diagram in the tri-articulated arch.

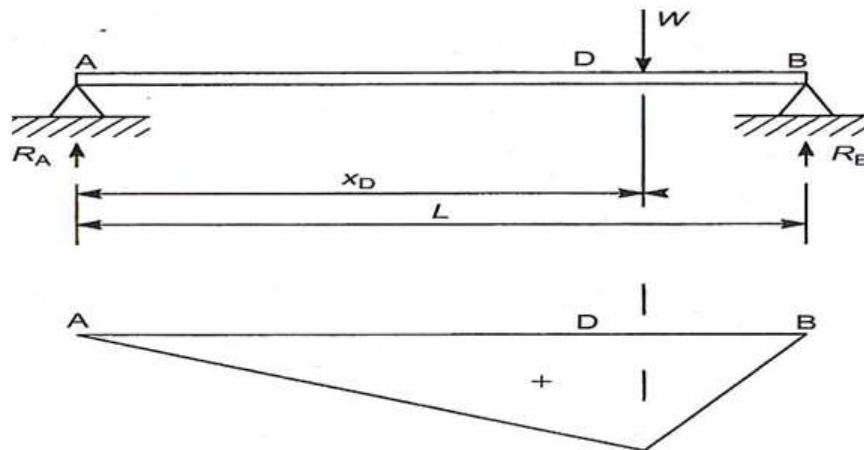


Figure (III.6b): Bending diagram in the corresponding beam, under vertical forces.

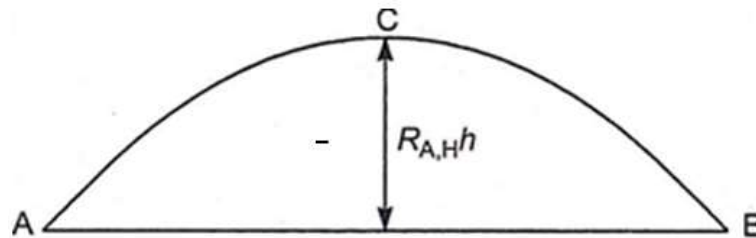


Figure (III.6c): Arch bending diagram due to horizontal reactions.

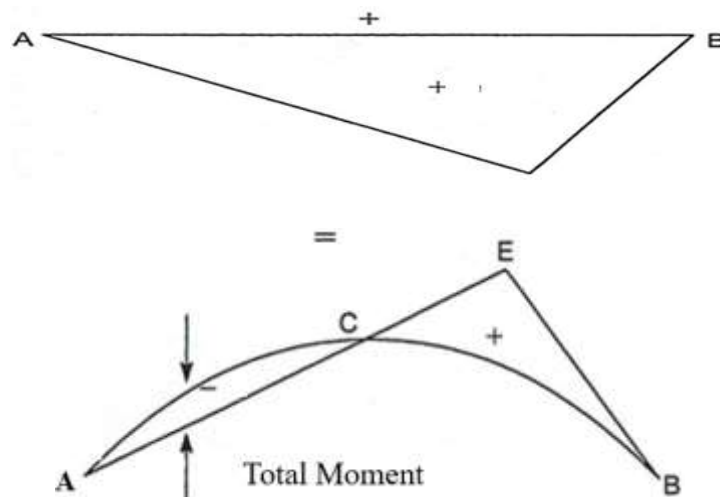


Figure (III.6d): Final bending diagram of the arch

III.3.2. Circular tri-articulated arch

Consider the semi-circular arch of (fig.III.7a) subjected to two loads (60kN, 100kN), whose positions are known since the equation of the arch is generally known. The third joint is assumed to be at the apex at point 'C'. The height of the

arch is 6m. **The supports at A and B are double; never single otherwise the arch may collapse.**

Let calculate the example below:

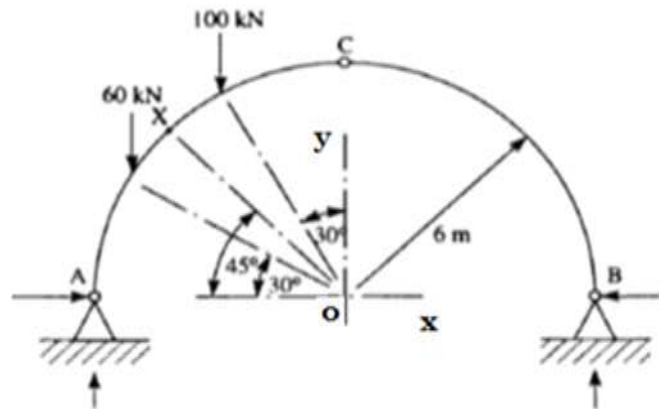


Figure (III.7a) : Calculation example of the normal force, the shear force and the bending moment at point 'x' of a semi-circular tri-articulated arch on the same levels supports.

The arch equation is written:

$$x^2 + y^2 = R^2 = 36$$

So :

$$y = \sqrt{36 - (6 * \sin 45)^2}$$

The sum of the moments with respect to point B:

$$A_y (12) - 60 (6\cos 30^\circ + 6) - 100 (6\sin 30^\circ + 6) = 0$$

$$A_y = 131\text{KN}$$

The vertical equilibrium of forces gives:

$$A_y + B_y - 60 + 100 = 0$$

By substitution

$$B_y = 29\text{KN}$$

Since no horizontal load exists:

$$A_x + B_x = 0$$

The moment to the right of C is zero, we write:

$$B_x * 6 - B_y * 6 = 0$$

$$A_x = B_x = 29\text{KN}$$

The normal force at point 'x' is obtained by the partial equilibrium in the direction tangent to the arch:

$$N_x = -A_y \cos 45^\circ - A_x \sin 45^\circ + 60 \cos 45^\circ$$

$$N_x = -70.7 \text{ kN}$$

The sign (-) indicates that the normal force in the arch is compression.

The shear force at point x is obtained by the equilibrium by part in the direction normal to the arch

$$V_x = A_y \sin 45^\circ - A_x \cos 45^\circ - 60 \sin 45^\circ$$

$$V_x = +29.7 \text{ kN}$$

Moment at point 'x' is written as:

$$M_x = A_y(6 - 6 \cos 45^\circ) - A_x * 6 \sin 45^\circ - 60(6 \cos 30^\circ - 6 \cos 45^\circ)$$

$$M_x = +50 \text{ kN.m}$$

Plotting the moment diagram in the semicircular tri-articulated arch

$$M(\text{arch}) = M(\text{equivalent rectilinear isostatic beam}) - A_x * y$$

$- A_x * y$: Horizontal reaction moment

M (simply supported beam); moment of the isostatic beam under vertical static loads (60kN and 100kN). The different diagrams are given in fig. (7b, c, d, e).

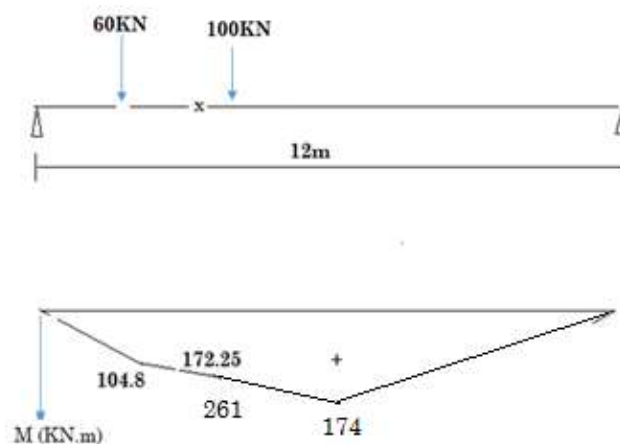


Figure. (III.7b): Moment of the equivalent rectilinear isostatic beam, under vertical forces.

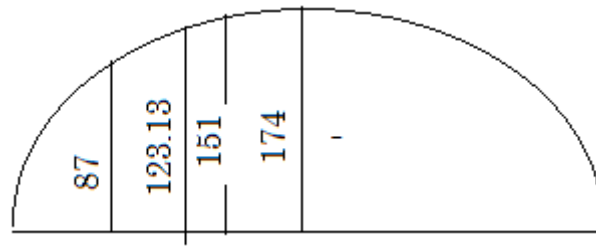


Figure (III.7c): Moment of the horizontal support reaction in (KN. m):

$$A_x * y = 29 * \sqrt{36 - x^2}$$

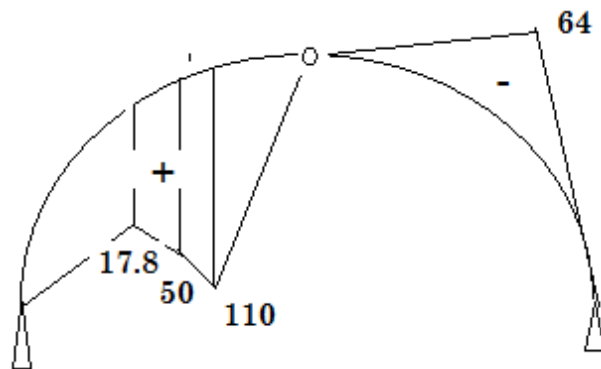


Figure (III.7d): Moment of the tri-articulated arch beam in (KN. m), superposition of the two diagrams in fig. (III.7b) and fig. (III.7c).

III.3.3. Tri-articulated parabolic arches loaded symmetrically

For the tri-articulated arch bridge of parabolic shape, subjected to a symmetrical load uniformly distributed on the horizontal (fig.III.8a), we can demonstrate by the following that only a normal compressive force can develop in the arch (M and V are zero). The two supports of the arch are here at the same level.

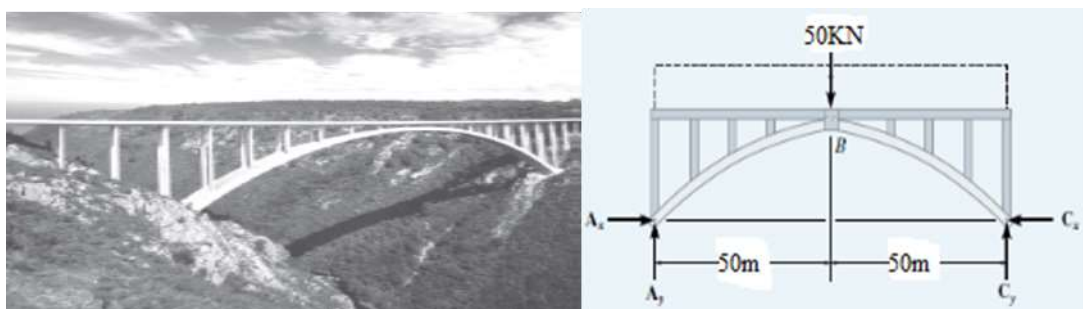
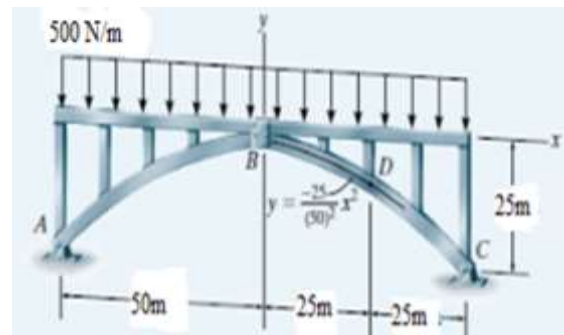


Figure (III.8a): (left) Tri-articulated arch bridge of parabolic shape, (right) calculation example

Figure (III.8b): Static diagram of the Arch example, subjected to a symmetrical load, uniformly distributed horizontally.



Global equilibrium of the arch gives:

$$\sum M_A = 0; \quad C_y * 100 - 50 (50) = 0$$

$$C_y = 25 \text{ KN}$$

Let us apply the equilibrium equations to part BC:

$$\sum M_B = 0; \quad -25 (25) + 25 (50) - C_x (25) = 0$$

$$C_x = 25 \text{ KN}$$

$$\sum F_x = 0$$

$$B_x = 25 \text{ KN}$$

$$\sum F_y = 0; \quad B_y - 25 + 25 = 0$$

$$B_y = 0 \text{ KN}$$

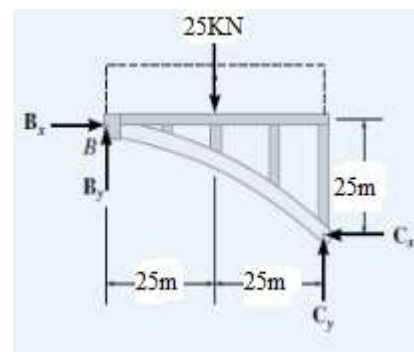


Figure (III.8c): BC static diagram.

Let the section of the arch taken at point D ($x=25\text{m}$) be the arrow $y = -25(25)^2/(50)^2 = -6.25\text{m}$. The rotation of the segment at point D ($x=25\text{m}$) is:

$$\tan \theta = \frac{dy}{dx} = \frac{-50}{(50)^2} x = -0.5$$

$$\theta = -26.6^\circ$$

Let us apply the equilibrium equations to BD:

$$\sum F_x = 0; \quad 25 - N_D \cos 26.6^\circ - V_D \sin 26.6^\circ = 0$$

$$\sum F_y = 0; \quad -12.5 + N_D \sin 26.6^\circ - V_D \cos 26.6^\circ = 0$$

$$\sum M_D = 0; \quad +M_D + 12.5 * 12.5 - 25 * 6.25 = 0$$

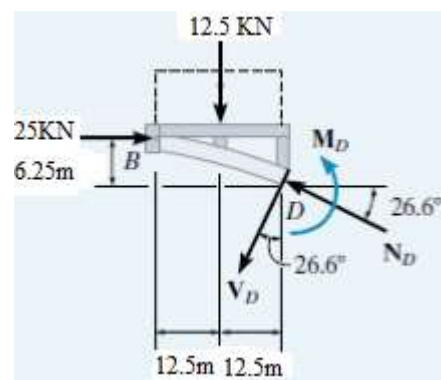


Figure (III.8d): BD static diagram

Finally:

$$N_D = 28.0 \text{ KN}$$

$$V_D = 0$$

$$M_D = 0$$

- ✓ If the arch has a shape different from the parabola or if the load is non-uniform, then the internal forces (V and M) will no longer be zero. We present some of these cases in the following.
- ✓ Suppose that a straight double-supported beam (in place of the arch $M=0$) supports this load of 500KN/m, then this beam must resist a maximum moment of $(500 \cdot 100^2 / 8 = 625 \text{ KN.m})$.

It is easier for a structure to resist a load through an internal axial compressive force (you just have to consider the possibility of buckling) than to resist the load through a bending moment. We conclude that **the parabolic arch shape is of great interest (compared to the rectilinear beam).**

III.3.4. Tri-articulated arch with tie rod

Ties join the two ends of the arch and largely take over the tensile forces. These structures can be metallic or concrete; they are suitable for many fields: road bridges, railway bridges, pedestrian bridges.



Figure (II.9): Arches with tie rods used for bridges. The deck is lifted by hanger bars, which transmit the loads to the arch.

For the tri-articulated arch with tie rod in the example below, we determine the forces in the bars CH and CB. The GF bar in broken lines does not develop any effort.

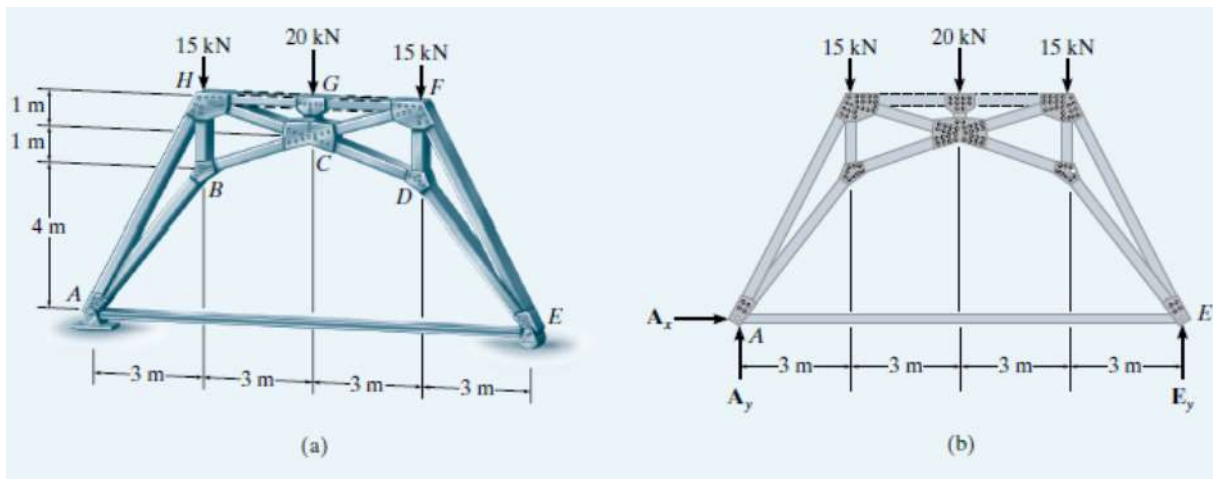


Figure (II.10a, b): (a) Example of an arch beam with tie rod, (b) static diagram of the arch with tie rod (in A double support, in B single support (roller)).

The support reactions are calculated from the global equilibrium of the arch:

$$\sum M_{/A} = 0; \quad E_y (12) - 15 (3) - 20 (6) - 15 (9) = 0$$

$$E_y = 25 \text{ KN}$$

$$\sum F_x = 0; \quad A_x = 0$$

$$\sum F_y = 0; \quad A_y - 15 - 20 - 15 + 25 = 0$$

$$A_y = 25 \text{ KN}$$

The equilibrium of the left part of the arch determines the two components of the force, which acts at point C. Let us first determine the force FAE.

$$\sum M_{/C} = 0; \quad F_{AE} (5) - 25 (6) + 15 (3) = 0$$

$$F_{AE} = 21.0 \text{ KN}$$

Moreover:

$$\sum F_x = 0; \quad -C_x + 21 = 0, \quad C_x = 21.0 \text{ KN}$$

$$\sum F_y = 0; \quad 25 - 15 - 20 + C_y = 0, \quad C_y = 10 \text{ KN}$$

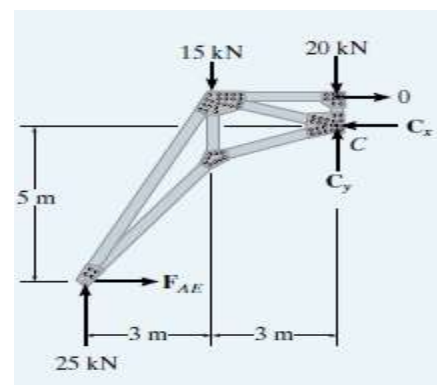


Figure (II.10c): AC static diagram

Let us now equilibrate the two nodes G and C to obtain the internal forces in the segments CH and CB:

Node G:

$$\sum F_y = 0; \quad F_{GC} - 20 = 0$$

$$F_{GC} = 20 \text{ kN}$$

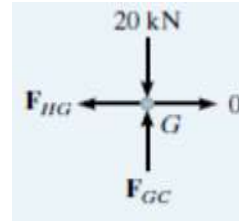


Figure (II.10d): Equilibrium of node 'G'.

Node C:

$$\sum F_x = 0; \quad F_{CB} \left(\frac{3}{\sqrt{10}} \right) - 21 - F_{CH} \left(\frac{3}{\sqrt{10}} \right) = 0$$

$$\sum F_y = 0; \quad F_{CB} \left(\frac{1}{\sqrt{10}} \right) + F_{CH} \left(\frac{1}{\sqrt{10}} \right) - 20 + 10 = 0$$

So:

$$F_{CB} = 26.9 \text{ kN}$$

$$F_{CH} = 4.74 \text{ kN}$$

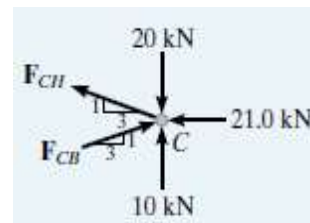


Figure (II.10e): Equilibrium of node 'C'.

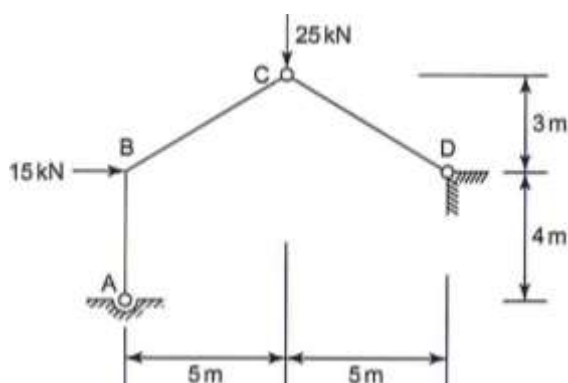
III.4. PROBLEMS

Problem 1: linear Arch

Consider the tri-articulated linear arch:

1. Calculate the support reactions in A and D.
2. Draw the moment diagram for members AB and BC.

N.A: $M_B = 14.1 \text{ KN.m}$

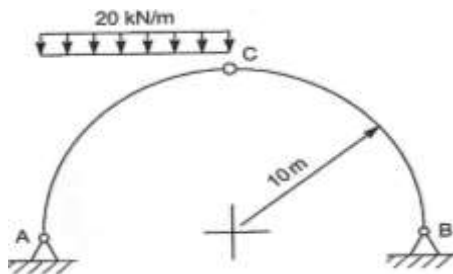


Pb.1

Problem 2: Circular Arch

Consider the tri-articulated semi-circular and half loaded arch. Determine the value of the Normal force, the shear force and the moment 5m from the left support.

N.A: $N = 68.3 \text{ KN}$, $V = 18.3 \text{ KN}$, $M = 67 \text{ KN.m}$

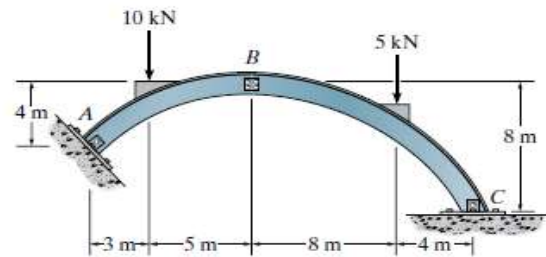


Pb.2

Problem 3: Arch of any geometry

1. Determine the horizontal and vertical components of the reactions at A and C of the tri-articulated arch, and the components of the forces at point B.

2. Draw the moment diagram.

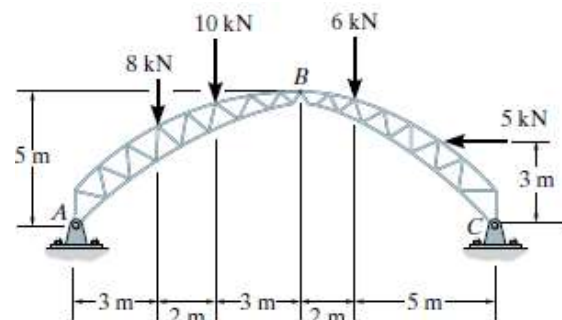


Pb.3

N.A: $R_{Ax} = 4.64 \text{ KN}$, $R_{Ay} = 8.57 \text{ KN}$, $R_{Cx} = 4.64 \text{ KN}$, $R_{Cy} = 6.43 \text{ KN}$, $B_x = 4.64 \text{ KN}$, $B_y = 1.43 \text{ KN}$

Problem 4: Arch of any geometry

Determine the magnitude of the forces at A, B and C of the tri-articulated arch-shaped truss roof at A, B and C.



Pb.4

N.A: $R_A = 19.9 \text{ KN}$, $F_B = 11.9 \text{ KN}$, $R_C = 10.4 \text{ KN}$

Problem 5: Parabolic Arch

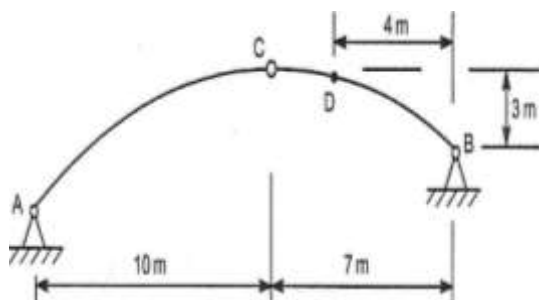
1. Given the arch as a tri-articulated parabolic arch, determine the differential equation of the arch, if it supports a load uniformly distributed

on the horizontal with an intensity of 40kN/m along the CB part.

2. Determine the horizontal and vertical components of the reactions at A and B, and calculate the moment at point D.

3. Draw the moment diagram.

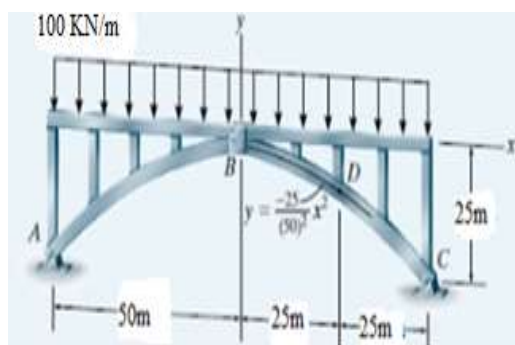
N.A: $y(x) = 0.075 x^2$, $R_{Ax} = 109.8 \text{ kN}$, $R_{Ay} = 82.35 \text{ kN}$, $R_{Bx} = 109.8 \text{ kN}$, $R_{By} = 197.6 \text{ kN}$, $M_D = 140.9 \text{ kN.m}$



Pb.5

Problem 6: Parabolic Arche

Determine the normal compressive force at point D, of the tri-articulated parabolic arch subjected to a load uniformly distributed on the horizontal of 100kN/m. The parabolic equation of the arch is $y(x) = \frac{-25}{(50)^2} x^2$.



Pb.6

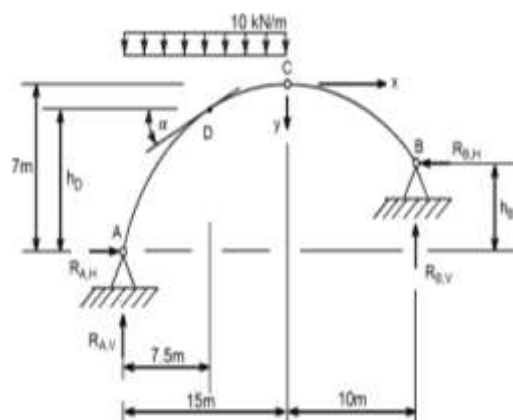
N.A: $N = 5.6 \text{ MN}$

Problem 7

Consider the parabolic arch beam, subjected to a load uniformly distributed on the horizontal and of intensity 10kN/m along AC.

Calculate the value of the normal force, the shear force and the moment at point D

N.A: $R_{Ax} = R_{Bx} = 95.5 \text{ kN}$, $R_{Ay} = 120.3 \text{ kN}$, $R_{By} = 29.7 \text{ kN}$, $h_B = 3.9 \text{ m}$, $h_D = 5.25 \text{ m}$, $\alpha = 25^\circ$, $N_D = 105.7 \text{ kN}$, $V_D = 0.7 \text{ kN}$, $M_D = 119.6 \text{ kN.m}$

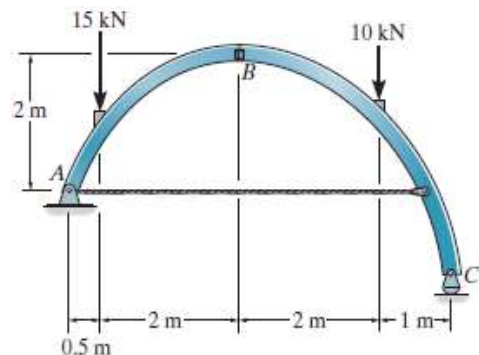


Pb.7

Problem 8: Arch with Tie Rod

1. Determine the reactions in A and C of the tri-articulated arch with tie rod.

2. Determine the tension in the tie rod



Pb.8

N.A : $R_{Ax} = 0$, $R_{Ay} = 15.5 \text{ kN}$, $R_C = 9.55 \text{ kN}$, $T = 4.32 \text{ kN}$